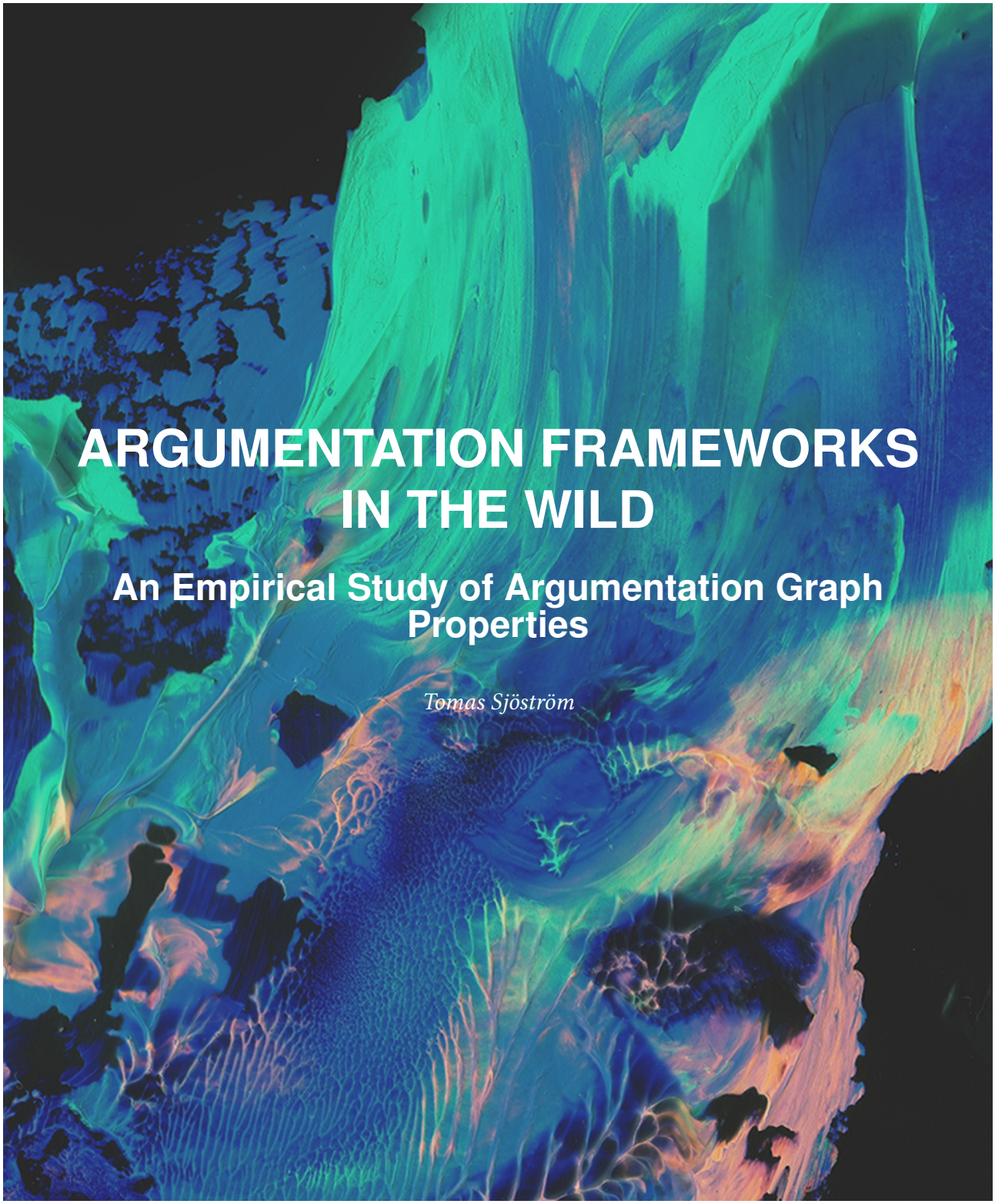
UMEÅ UNIVERSITY



ARGUMENTATION FRAMEWORKS IN THE WILD

An Empirical Study of Argumentation Graph Properties

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Bachelor Thesis, 15 hp/credits
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Abstract

This thesis does an empirical analysis of explicit generated argumentation graphs from a real-world dataset. Thereby, the thesis expands existing research, developing a procedure to extract and analyze the underlying structure of generated argumentation graphs and applying this procedure to a real-world dataset based on movie ratings. Studying the structure of argumentation graphs generated from real data allows for the validation and extension of theory-oriented work on computational argumentation.

The graphs studied in the thesis are based on existing work in which computational argumentation is applied to a recommendation system problem. This thesis looks into the underlying structure of 23,210 generated graphs and investigates assumptions made in theory.

The thesis finds that the dataset used to generate the graphs generates unbalanced graphs where some parts of the graphs can be empty. It accordingly finds that the conceptual design of the argumentation-based system should be expanded to take this into account when weighting influence factors during the generating of argumentation graphs. The empty parts of the graphs occur because the underlying structure is relying on the distribution of common elements between movies that are by themselves unevenly distributed. These structures are *genre*, *actors*, and *directors*. Many movies have genres in common while some have common actors but a significant number of movies lack common directors leading to empty sub-graphs.

Keywords: Formal Argumentation – Computational Argumentation – Recommendation Systems – Movies – Films – Graph Structures

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1 Introduction

Computational Argumentation (CA) is a long-running line of research focusing on reasoning about arguments and their relationships, where the arguments are modeled with different relationships between them. There exist different approaches such as Quantitative Bipolar Argumentation (QBA). A *Quantitative Bipolar Argumentation Framework* (QBAF) is modeled as a graph with nodes as arguments and the relationships between the arguments as edges representing attackers and supporters [1]. The strength of different arguments is given by the weight of the nodes. Inferences are drawn by updating *initial strengths* to *final strengths*, considering graph topology.

Another similar argumentation framework with attackers and supporters is described in [7]. The framework is used for a recommendation system that can for instance be used to recommend movies where the final recommendation of an item is based on a set of recommendations of other items.

Many papers corresponding to the research about argumentation frameworks are theoretical with limited studies of how the actual graphs based on real data is structured. The paper [7] implicitly generates graphs and measures high-level information such as final strength. The paper [7] has not explicitly generated the graphs used from the argumentation framework leaving a research gap where the explicit structure of the graphs generated is not understood. This leads to the research question in the next section where the structure is investigated.

1.1 Research Question

What properties can be investigated from the argumentation frameworks based on [7] if the graphs are explicitly extracted and their properties investigated?

1. Are the graphs cyclic or acyclic?
2. How are the graph sizes distributed?
3. What is the distribution of attackers and supporters?
4. How are the argument strengths distributed across the graph?

1.2 Contribution

The thesis contributes to existing research, expanding the work in [7] by developing a procedure to extract and analyze the structure of argumentation graphs generated and studied from a real-world dataset. This gives a deeper understanding in the structure of the graphs generated and validation for assumptions made in theory. The work also contributes with empirical methods to evaluate the structure of these graphs.

2 Related Work

This section gives an overview of the related works for the thesis corresponding to Computational Argumentation (CA).

2.1 Quantitative Bipolar Argumentation

There exist several different argumentation frameworks. One of those is *quantitative* argumentation that is a recent field being researched. Here the graphs consist of nodes representing arguments that can either attack or support each other while strengthen or weaken arguments. These graphs usually starts with initial weights on the the nodes. The final strengths of the graphs are then computed using argumentation semantics that infer the strength [3] based on other nodes and the initial node strength.

Commonly studies rely on synthetic generated data [6, 8, 4]. Even when quantitative CA methods are applied to real-world datasets, little attention is paid to the properties of the argumentation graphs that these datasets give rise to.

2.2 Recommendation Systems

A recommendation system is a system used to recommend an item. One area where argumentation can be used is in a recommendation system where earlier ratings can be be seen as argument that either supports or attacks a new estimated rating. There exist many different approaches such as in [2] that is based on logic programming or a hybrid rule-based approach in [9]. There exist complex models [10] that are harder to explain and models that focus on explainability.

A less complex and easier to explain recommendation system is the argumentation framework described in [7]. It describes one way to explicitly construct an argumentation framework that is similar to QBAF.

The framework in [7] estimates the final strength of a new rating as weighted averages of known ratings. The framework has the arguments as nodes and the inference between the nodes are based on the weighted average of the strength of previous nodes. The strengths of the nodes are between -1 to 1 where the positive values are supporters while the negative values are attackers.

2.3 Similarities With QBAF

Both QBAF and the framework [7] uses arguments for the nodes and attackers and supporters. The way the attackers and supporters is determined in different ways. In the QBAF they are set at the start while in the argumentation system from [7] they are determined during the propagation of the data through the framework. Both have quantitative strengths for each argument nodes.

2.4 Research gap

The paper [7] implicitly construct the graphs and measure high-level values such us the final rating used to measure the accuracy of the argumentation framework. These graphs has not explicitly been created or studied. Doing this would allow for the measurement of low level properties like the distribution of attackers and supporters. This leaves a research-gap for this thesis to empirical explore explicitly generated graphs generated from the real data and to look at different properties of these argumentation graphs in order to understand the structure of the argumentation graphs. Due to the framework being close to but not explicitly being a QBAF there is also room for looking into how to translate the graphs to QBAF for in order to measure properties of the explicit graph.

3 Background

This section contains an explanation of the existing argumentation framework applied to the recommendation system this thesis expands on.

3.1 Quantitative Bipolar Argumentation Framework

Quantitative bipolar argumentation frameworks describe graphs that show the relationship between different arguments represented as nodes [5]. The quantitative part is that each node has a value and the bipolar part is that different arguments can either support or attack other arguments. The QBAFs have initial strength for each node,

3.2 Argumentation Framework

The framework for the graph generation used in this thesis is based on the framework from [7]. An example of the argumentation framework can be seen in Figure 1 where argument A with the positive strength of 1 supports argument C while argument B attacks argument C with the negative strength of -1. The final strength is the weighted average of the arguments.

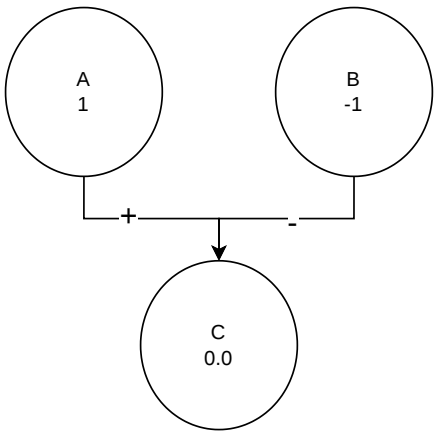


Figure 1: Shows the relationship between three arguments where argument A supported argument C and B attacks argument C.

3.3 Recommendation Systems

The argumentation framework from [7] that is used to describe a type of recommendation system where users u recommends different items i based on how they have recommend

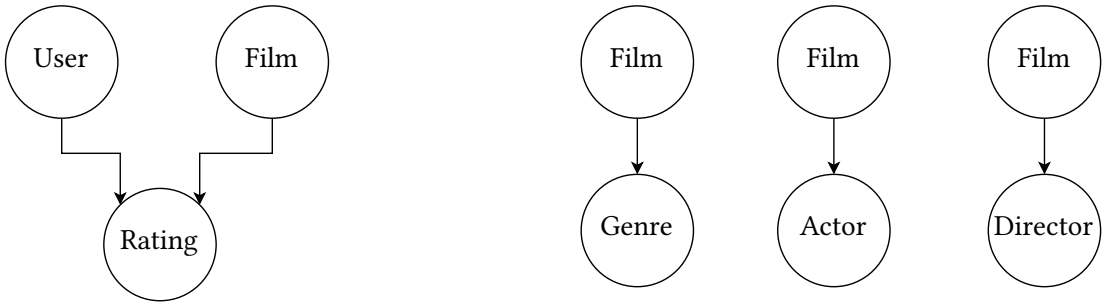


Figure 2: Shows an example of the relationship between the data used in the argumentation framework described in [7]. The users have rated some films where the films share the the same genres, actors and directors.

some of the items before. The items have different aspects A in-common that are not in the set of items.

One example used for the framework in [7] is films where there the users u are the viewers that rates different items i that in this case are films. The aspects A that connect different films would be for instance different genres like action or comedy. Each A has one aspect type T that it can share with other aspects. In the case of films this can be genre g, actor a or director d. The relationships between the u, i and A in the data that the graphs are build on can be seen in Figure 2.

The goal of the recommendation system is to give a rating for the user and item pair. The ratings ratings are as in [7] either the exact user's rating where those are available or the a weighted average of the estimation of the aspects as in Equation (3.1) when $R_A^u(i)$ is defined and where $P_A^u(i)$ is the final strength, $R_A^u(i)$ is the rating of Item i. In the film case it would be the weighted average of the estimated rating for the genre, actors and director of the current film that is being rated. In case the rating for the user item pair does not exist the user u rating P of item i with initial ratings R can according to Definition 3 in [7] be written as in Equation (3.1) and Equation (3.2) where $\mu_t^u \in]0, 1]$ is a constant determining the importance of the T, t is one type of the set of all the types T of $I's$, c is the predicted rating as described in [7].

$$P_A^u(i) = R_A^u(i) \tag{3.1}$$

$$P_A^u(i) = \frac{\sum_{t \in T} \left(\mu_t^u c_t^{u,i} \right)}{\sum_{t \in T} \left(\mu_t^u \right)} \tag{3.2}$$

The framework estimates the rating for each aspect based on the data from Figure 2. This is done in a series of steps. Each category of aspect (for instance genre, actor or director) is averaged based on different aspects that are associated with the item investigated. For instance if the film that is to be rated is an action comedy the framework would first estimate a rating for the the genre comedy and action separately. Figure 3 shows how one subgenre is estimated based on the films in the same subgenre that the user has seen. Similar is done for actors and directors.

A set of similar user is used to complement the ratings from the user. Similar users are given based on the cosine similarity [7] based on how the users have rated the movies. Each similar user is calculated like the user and then average in to one rating as in Figure 4.

The rating for each aspect A is then computed as the weighted average between the main

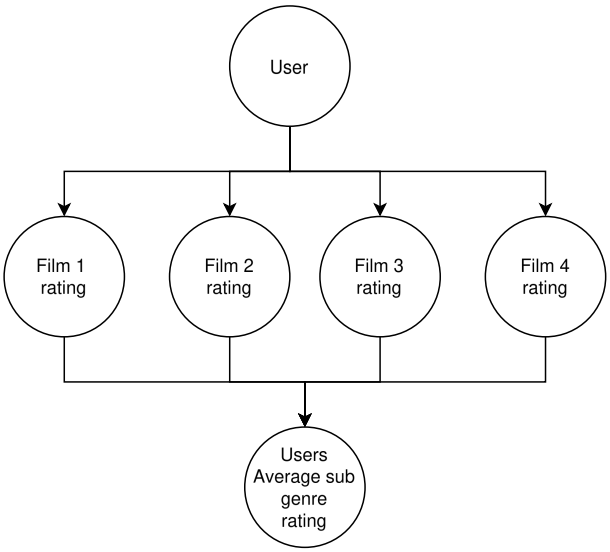


Figure 3: Shows how each genre is being estimated as the average of the movies in the genre that the user have rated.

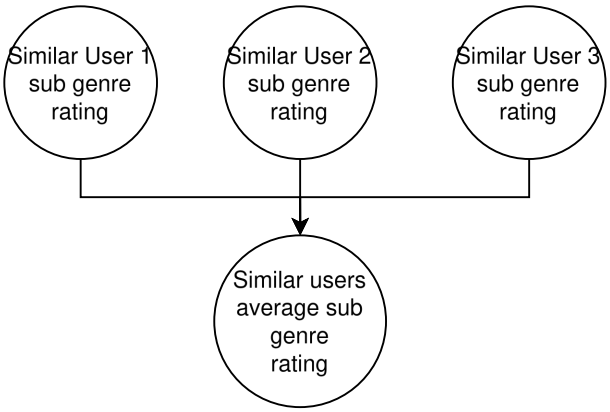


Figure 4: Shows how similar users are being averaged into an estimation of the their rating for the genre.

user rating and rating from the similar users. Figure 5 shows a visualization of the weighted average between the user and similar user for the aspect genre with the weight given by MUR. Each sub genre is then average together to the full rating for the genre as in Figure 6.

The final film rating is then computed as a weighted average of the aspects A. For the case of films the weights can be given as the weight for the genre as MUG, the weight for the actors as MUA, the weight for the director as MUD and the weight for the user compare to the similar users as MUR. A visualization of the computation of the final strength can be seen in Figure 7.

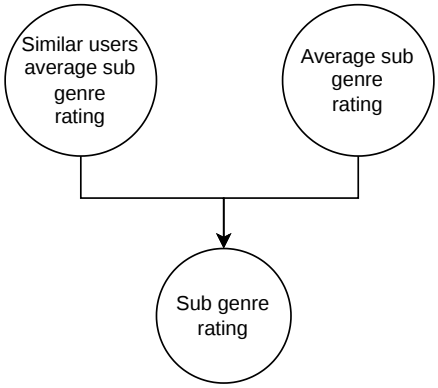


Figure 5: Shows how the rating of the user and the similar user is averaged with the weight MUR.

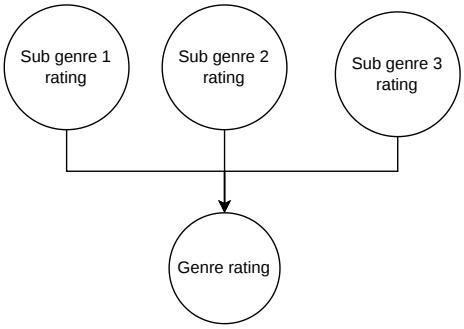


Figure 6: Shows how each sub genre connected to the original movie is averaged together.

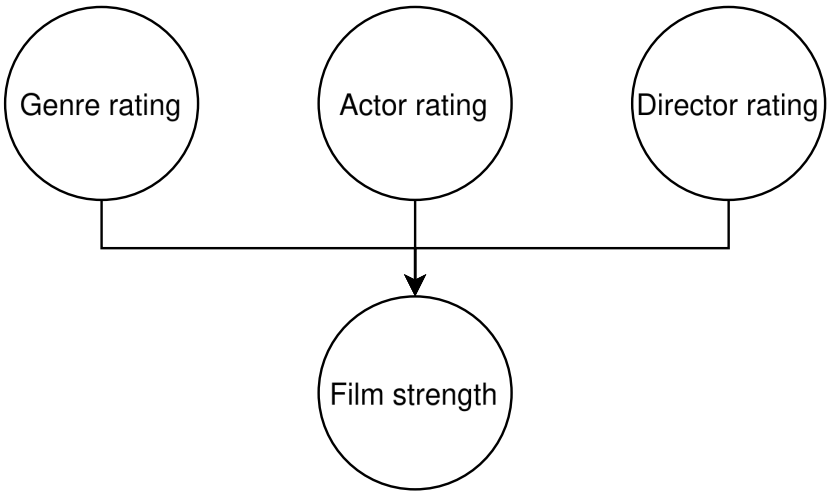


Figure 7: Calculation of final movie strengths: the weights average of the estimation of the genres, actors and directors use the weights MUG MUA and MUD.

4 Method

This section describes how the graphs from [7] are extracted and analyzed.

4.1 Problem Formulation

The objective of the thesis is to empirically study the properties of the graphs generated by the argumentation framework described in [7] and investigate properties of the graphs. This is achieved by generating graphs from an available dataset from [7]. The graphs is generated using code that is based the code from [7]. The graphs are studied using an expanded version of the tool developed for studying QBAFs [11] to calculate the distributions of attackers and supporters. The graphs are generated from the datasets accompanying the code in [7]. The code has been expanded in order to be able to extract the features from the argumentation graphs. The study is constrained by the framework used in the [7] and by that the analysis has only been run on a movie dataset.

4.2 System Architecture

All graphs are explicitly generated from [7] by extracting each node's argumentation strength as well as there corresponding edges into a separate data-structure that is then stored in PKL files separated by each user. The PKL files containing all the graphs connected to the movie ratings for each user are later loaded into the analysis tool where the structure of the graphs are analyzed and plotted. Figure 8 shows a high level diagram of how the graphs are being extracted and analyzed.

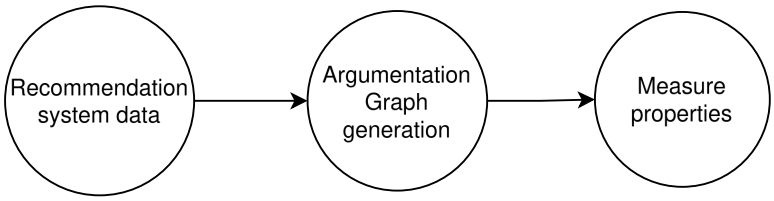


Figure 8: A high-level overview of how the the graphs are being extracted.

4.3 Model / Algorithm Description

The section contains the methods for generating and analyzing the graphs.

4.3.1 Graph Generation

As the graphs have been implicitly generated using nested for loops in [7] they have been extracted by adding a graph generation to where the argument strengths are generated. Each

corresponding argument strength is added to a node in the explicit graph structure connected with the corresponding edges to build a structure corresponding to Figure 9.

4.3.2 Analysis

The analysis is done using the tool [11] extended to handle the graphs from the recommendation framework. The analysis is done by traversing the generated graphs using a breath first search and summing the corresponding attacker or supporters for each graph. An argument is considered an attacker if their strength is negative and a supporter if the strength is strictly positive. The number of attackers and supporters are then statistically analyzed using mean, median, standard deviation and threshold for the minimum, maximum and where 25% as well as 75% of the values are distributed.

4.4 Experimental Setup

The code for extracting the graphs and the code for the analysis are openly available[1]. The code used for extracting the graphs is written in Python using Jupyterlab with the library sklearn, numpy and pandas. The code used for analysis uses CPython (Python with C API).

4.4.1 Evaluation Methodology

The properties measured from each graphs are the the number of edges corresponding to the size of the graphs. The total number of attackers and supporters used to evaluate the final rating as well as the number of attackers and supporters used evaluate the rating for the genre, actor and director. The number of initial arguments attacking or supporting used for the estimating the genre, actor and director strength are also counted. The final strength as well as the strength for the genres, actors and directors are also measured.

The measured data is then aggregated over all generated graphs and measured statistically using the mean, median, standard deviation, the minium value, the limit for 25% and 75% of the values and the maximum value.

4.5 Datasets

The dataset used is included in [7] that describe a recommendation system for movie is studied. The dataset has 943 users and the dataset is generated 2020 with data from [2]. The movies are rated in the dataset ranges from 1 to 5 that is rescale to -1 to 1 during the graph generation.

4.6 Limitations Of Methodology

The study is an empirical study limited by the choice of the original framework that limits the structure of the graphs that can be created. For instance the graphs that are made becomes acyclic by design. It's also limited by the datasets used. To expand beyond a recommendation system another framework would need to be used to generate the initial graph. Another limitation is also that only one dataset is used limiting the result to the biases in the dataset.

[1] https://github.com/tomassjostrom1/KR2020-Aspect-Item-Recommender-System, https://github.com/tomassjostrom1/Quantitative-Bipolar-Argumentation

[2] https://www.themoviedb.org/movie/

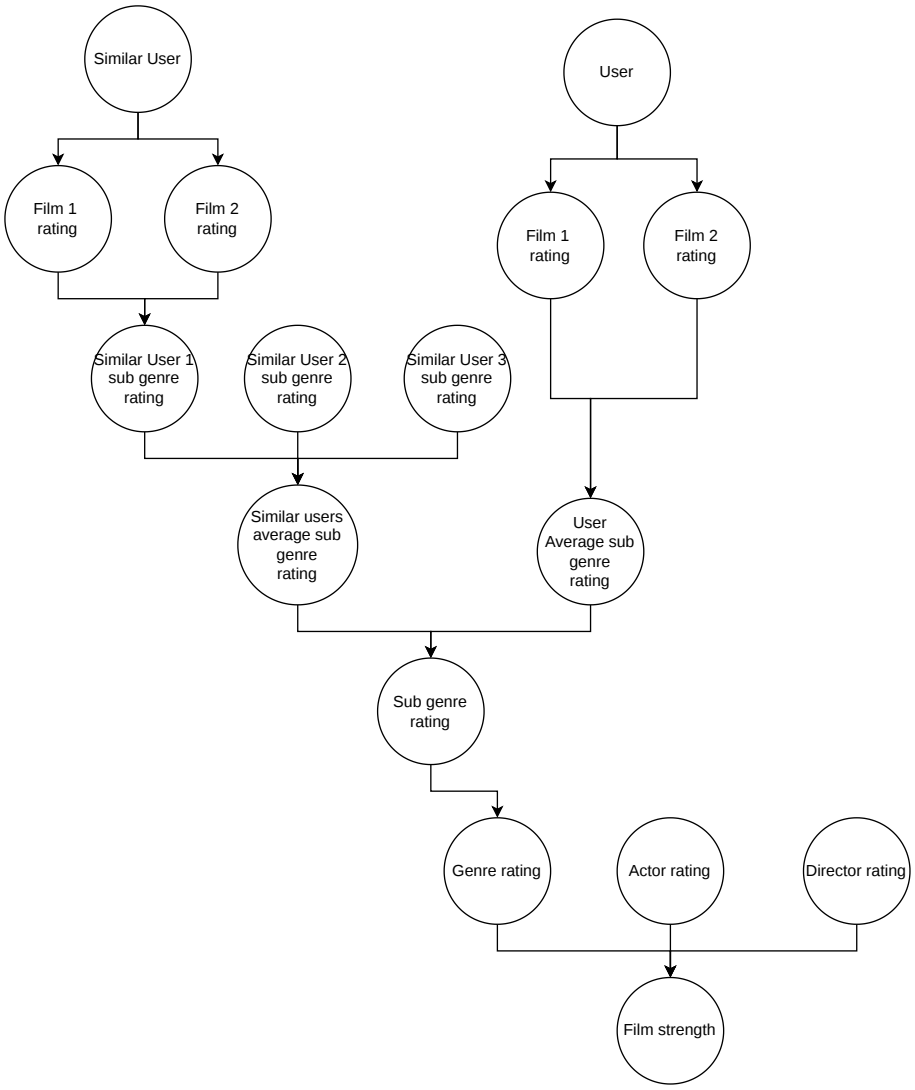


Figure 9: A schematic overview for how the graphs are generated in different layers. Where First the users ratings are considers and average together for each subgenre, for instance action or comedy. Then the rating for each genre is an average of the genres that the movie has. The corresponding (For easier readability the subgraphs for the actors and directors has been left out) thing is done for the actor and director. The final film strength is then the weighted average of the strength corresponding to the genres, actors and director.

The study is also limited be being done on a framework [7] that is not explicitly a QBAF. This means that the result is in first hand valid for argumentation graphs following the [7] framework and limits the generalizations that can be done for QBAFs.

5 Results

For all examples below the results were generated with the parameters MUR = 0.1, MUG = 0.6, MUA = 0.1, MUD = 0.1 and 20 similar users. Totally 23,210 graphs where generated for 943 users.

5.1 Graph examples

Figure 10 shows a visualization of the explicit graph generated for user x48 estimated rating for the movie *Mrs. Brown Her Majesty*. The ratings for item is in the range -1 to 1 where a strictly positive rating is a recommendation and a negative rating is not a recommendation. How much the movie is recommended or not is determined by the size of the strength. In this case the rating is positive and the movie is recommended. This is due to the argumentation system identifies that the movies, actors and director corresponding to the movie all get positive ratings.

This is unlike in Figure 11 that shows the movie getting a negative rating for the movie *Rocket Man* (1997). This is due to argumentation system estimating the rating for the genre, actors and directors to be negative.

Another form of graph structure can be seen in Figure 12 showing the rating for *Willy Wonka and the Chocolate Factory* (1971) corresponding to user *x876*. As can be seen in the graph there is no information to estimate the rating for the actors and director leading to both ratings being set to 0; only the genre rating causes the final rating to be different from 0.

5.2 Distribution Of The Initial Layer

From Table 1 it is possible to see that the number of attacker and supporters from the I's used in the initial layer to estimate each A. The sum of the number of attackers and supporters in the initial layer (corresponding to the nodes to the right in Figures 10 to 11) can be found under the combined line for each A.

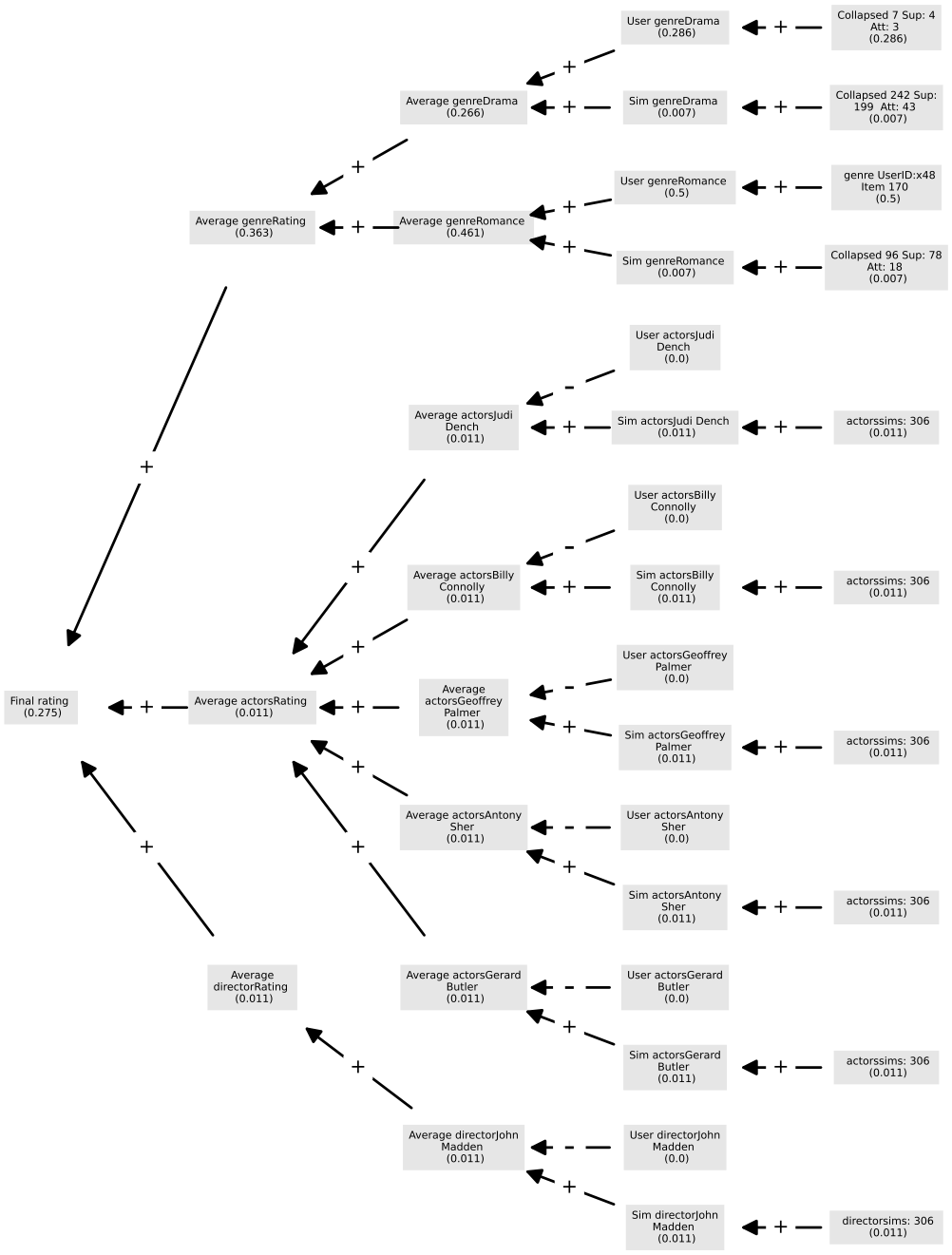


Figure 10: Show the explicit argument graph for the user x48 rating for the movie 306 corresponding to Her Majesty, Mrs. Brown with the weights MUR = 0.1, MUG = 0.6, MUA = 0.1, MUD = 0.1 and 20 similar users.

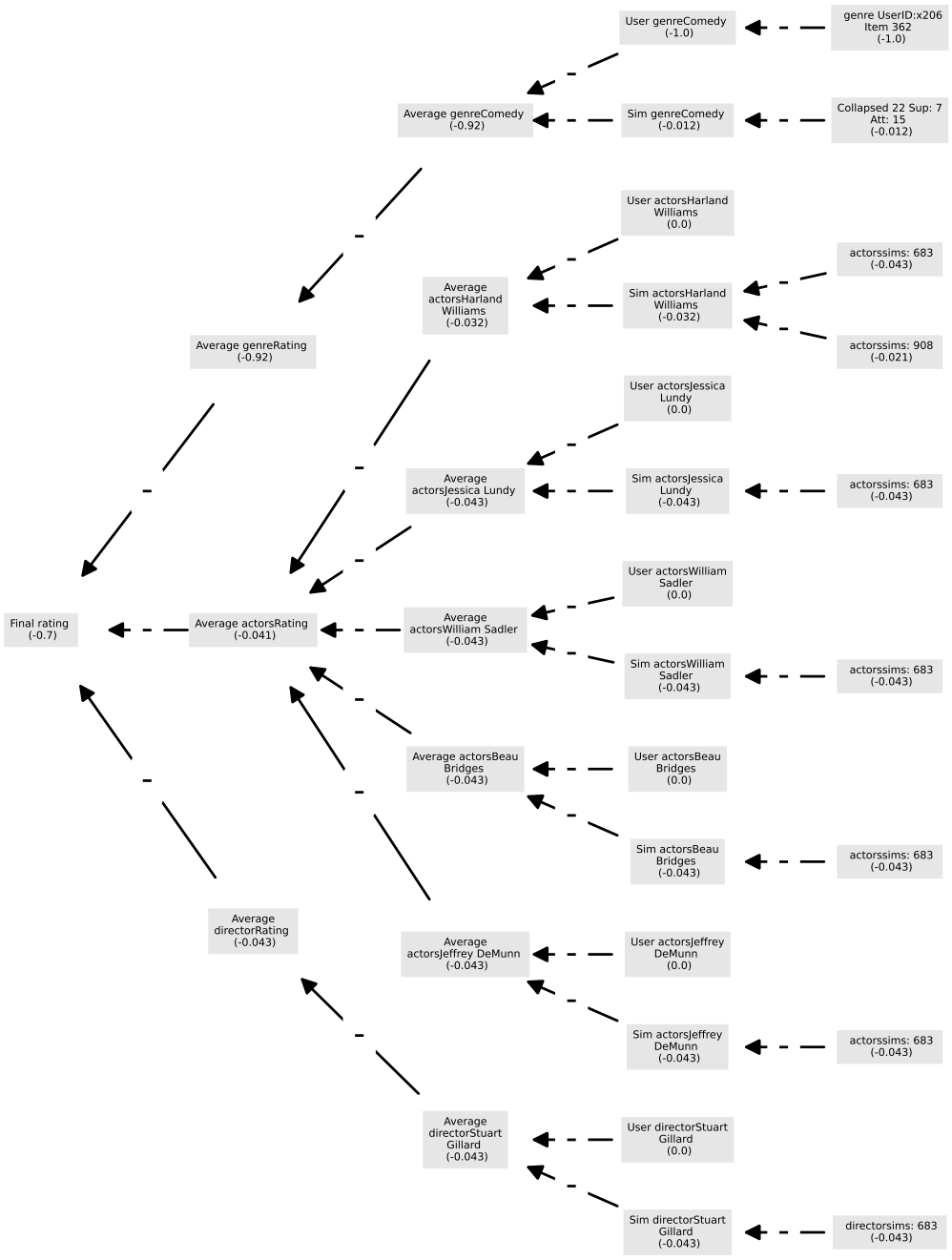


Figure 11: Show the explicit argument graph for the user x206 rating for the movie 683 corresponding to Rocket Man (1997) with the weights MUR = 0.1, MUG = 0.6, MUA = 0.1, MUD = 0.1 and 20 similar users.

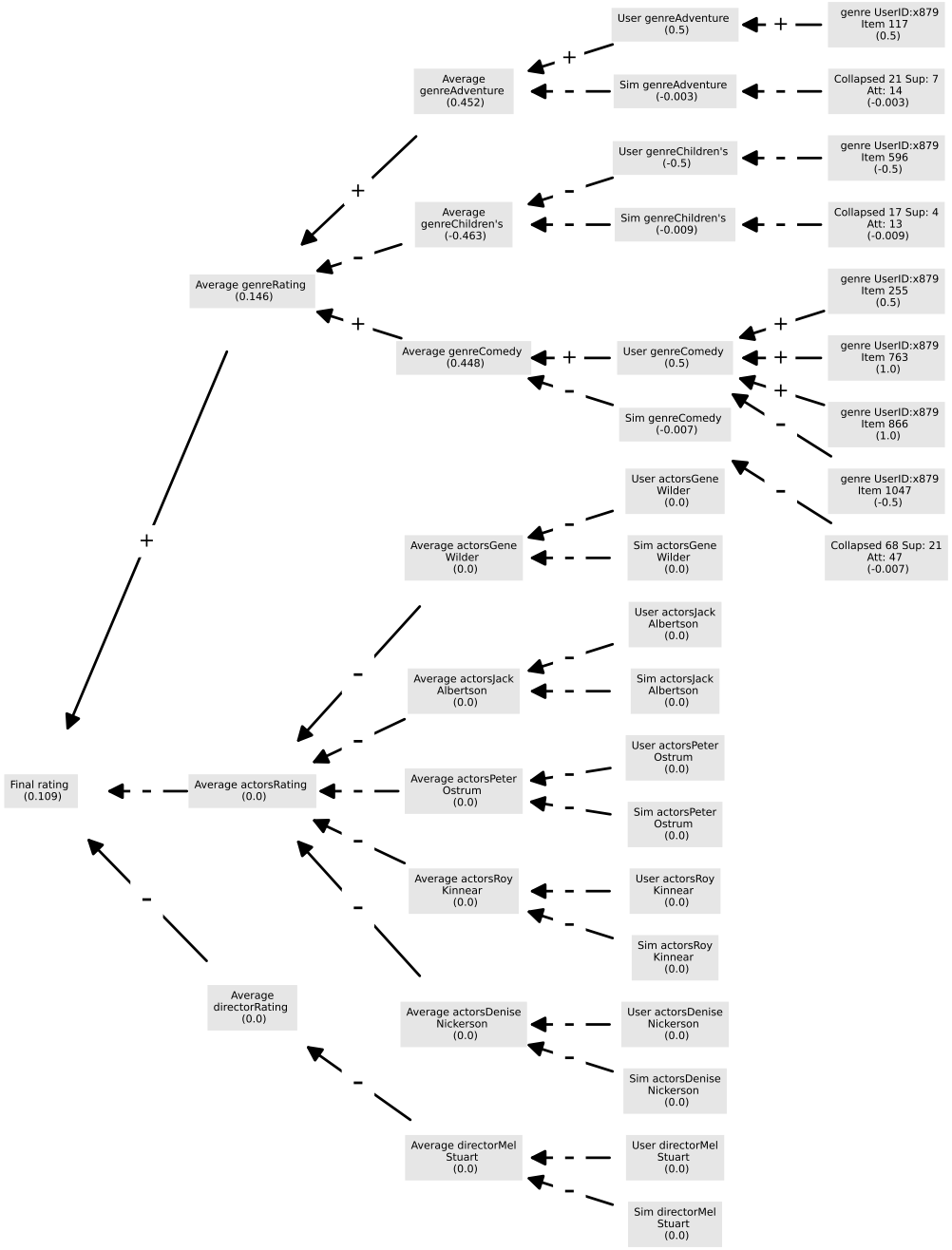


Figure 12: Show the explicit argument graph for the user x876 rating for the movie 683 corresponding to The Spitfire Grill (1996) with the weights MUR = 0.1, MUG = 0.6, MUA = 0.1, MUD = 0.1 and 20 similar users.

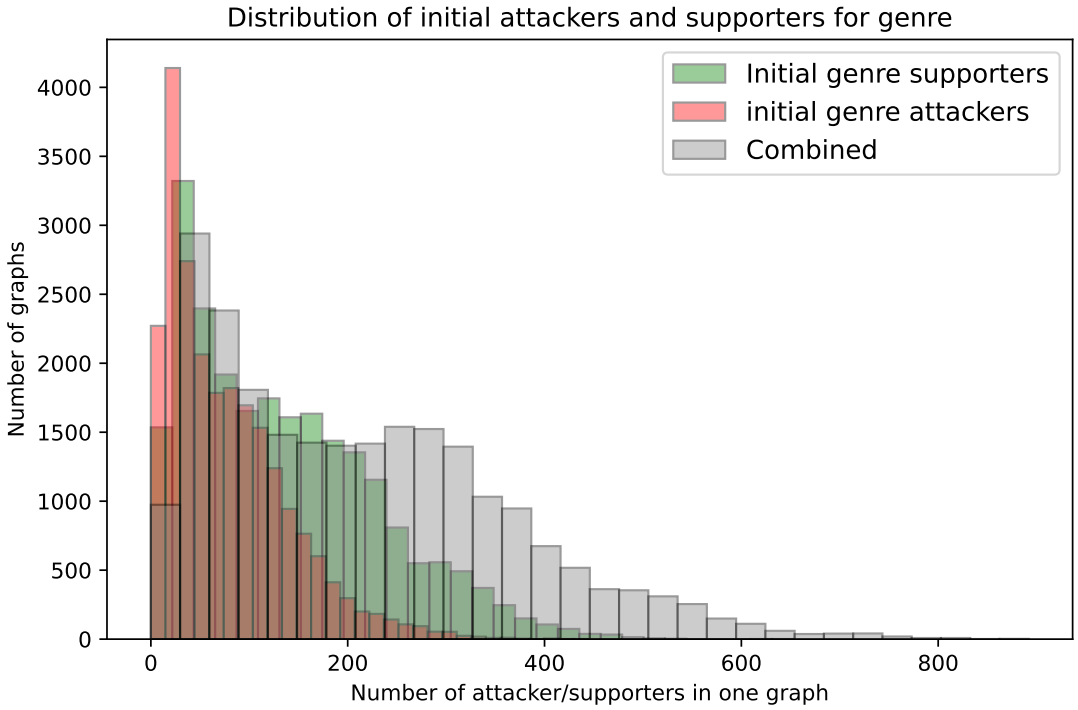


Figure 13: The distribution of attackers(strictly positive ratings) and supporter(negative rating) for the initial layer of items contributing to the ratings for the genres belonging to the movie.

Table 1 The distribution of attackers and supporter for the initial layer of items containing known ratings of movies by users used in the argumentation framework. The ratings are split into those that contribute to the ratings for the genre, those that contributes to the actors and those that contribute to the ratings for the directors. The contribution item is considered a supporter if its contribution is strictly positive and a attacker if it is negative.

Item	Mean	Median	Standard deviation	Min	25%	75%	Max
Genre supporter	135.17	118.00	96.70	0.00	51.00	198.00	653.00
Genre attacker	77.06	63.00	61.20	0.00	27.00	112.00	444.00
Genre combined	212.23	191.00	148.48	0.00	82.00	308.00	892.00
Actor supporter	6.96	6.00	5.20	0.00	3.00	10.00	36.00
Actor attacker	2.41	1.00	3.09	0.00	0.00	4.00	28.00
Actor combined	9.36	8.00	7.12	0.00	4.00	13.00	43.00
Director supporter	1.69	1.00	2.22	0.00	0.00	2.00	25.00
Director attacker	0.20	0.00	0.61	0.00	0.00	0.00	23.00
Director combined	1.89	1.00	2.46	0.00	0.00	2.00	25.00

Figure 13 shows the distribution of attackers and supporters among all graphs used to estimate the genre A for the I's used in the initial layer. Corresponding distribution of the attackers and supporters for the I's used to estimate the rating for the actor A can be seen in Figure 14. The distribution of attackers and supporters in the initial layer corresponding to the I's influencing the in rating for the directors is in Figure 15.

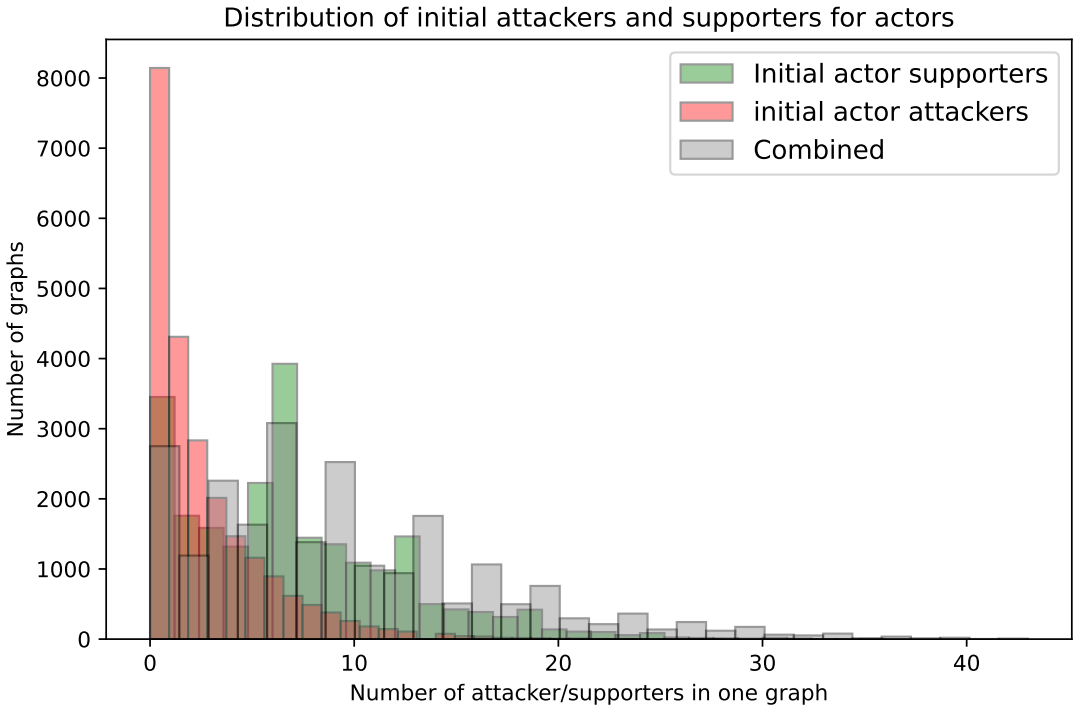


Figure 14: The distribution of attackers(strictly positive ratings) and supporter(negative rating) for the initial layer of items contributing to the rating for the actors belonging to the movies.

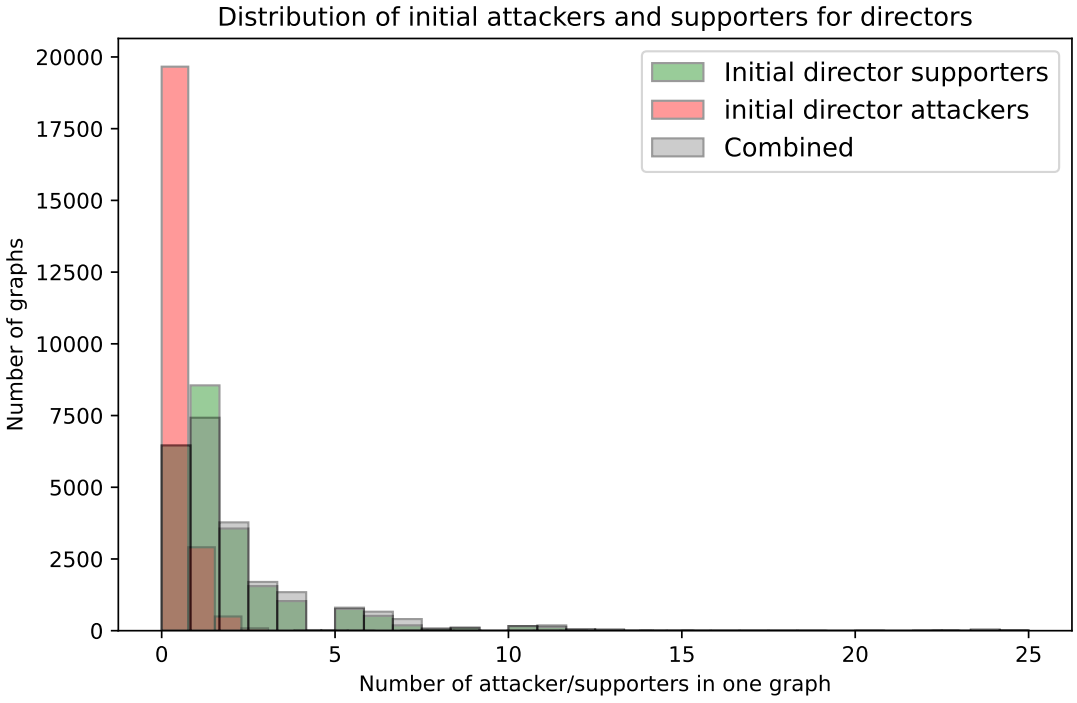


Figure 15: The distribution of attackers(strictly positive ratings) and supporter(negative rating) for the initial layer of items contributing to the rating for the directors belonging to the movies.

5.3 Distribution Of Graph Sizes

The distribution of the graphs number of edges can be seen in Table 2 with the corresponding distribution shown in Figure 16.

Table 2 Distribution of the number of edges for the graphs generated for the argumentation framework graphs counted from the final strength and for the corresponding genre, actor and director argument.

Item	Mean	Median	Standard deviation	Min	25%	75%	Max
Full graph	252.54	233.00	154.01	14.00	117.00	353.00	958.00
Genre	220.01	198.00	149.50	4.00	90.00	316.00	902.00
Actor	25.32	24.00	7.19	4.00	20.00	29.00	59.00
Director	6.22	5.00	4.62	4.00	4.00	7.00	62.00

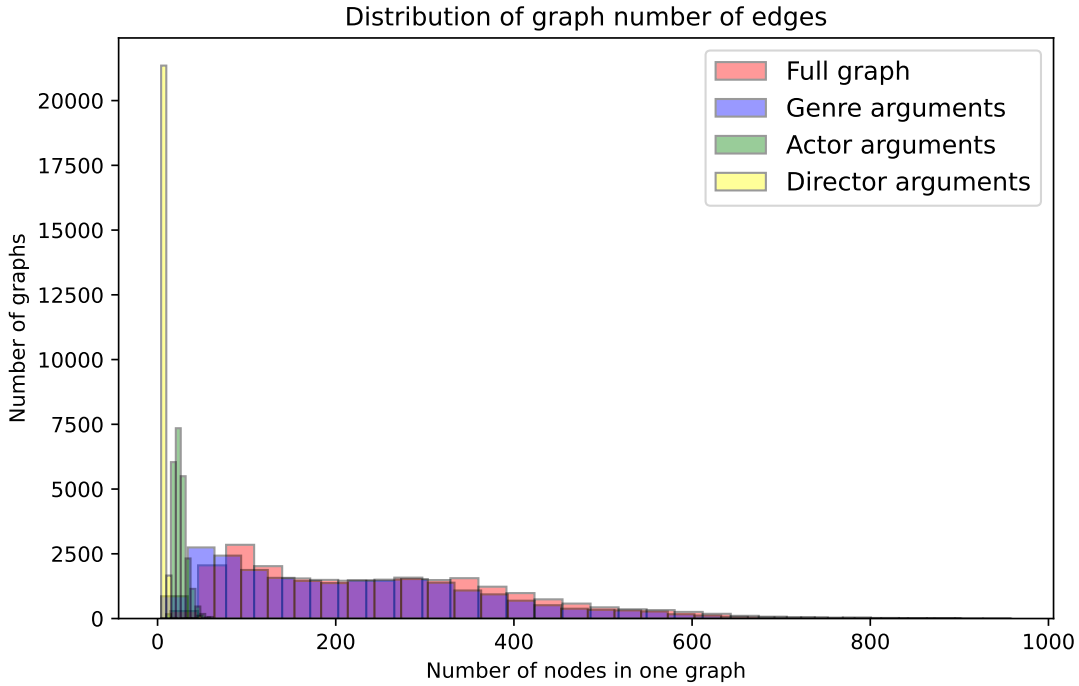


Figure 16: Distribution of the number of edges for the graphs generated for the argumentation framework graphs.

5.4 Distribution Of Indirect Attackers And Supporters

The statistics for the the number of attackers, supporters and there sum for the different parts of the graph corresponding to As genre, actors and directors can be seen in Table 3. The distribution of for the attackers and the supporter for the genre part can be seen in Figure 17. The corresponding distribution for the actors can be seen in Figure 18 and for the directors can be seen in Figure 19.

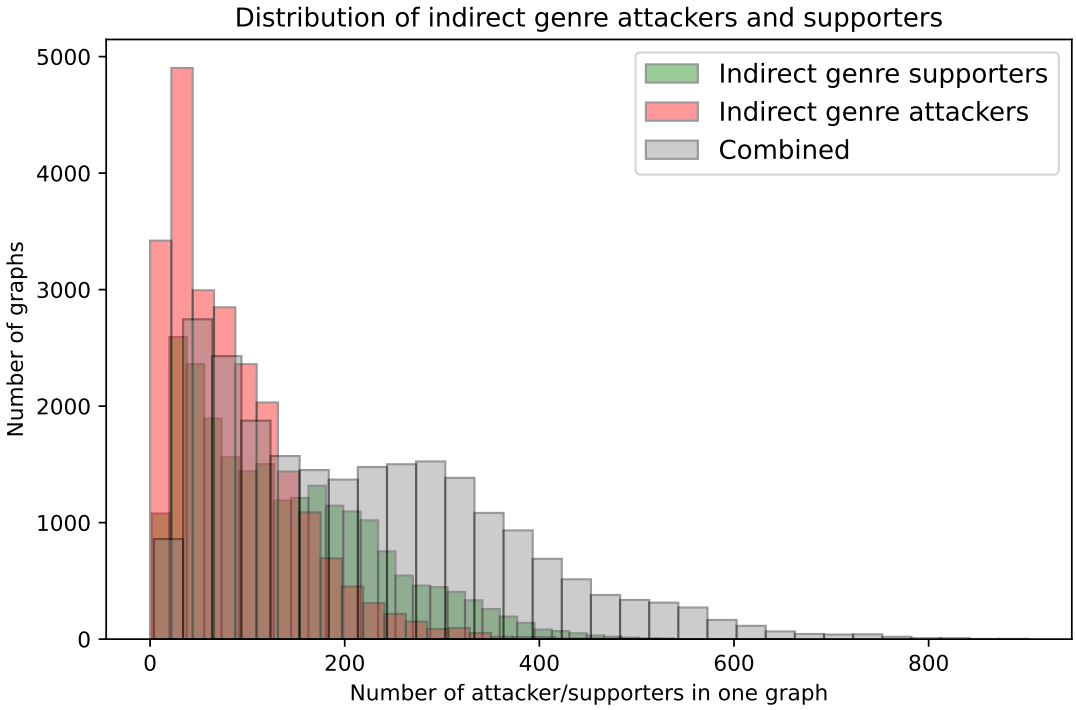


Figure 17: Shows the distribution of the indirect attackers and supporters contributing to the rating for the genres. The contributing item is considered a supporter if its contribution is strictly positive and an attacker if it is negative.

Table 3 The distribution of attackers and supporter for the subgraphs originating from different ratings aggregate from all 23,210 graphs. The genre attacker and supporters only considers attackers and supporters that contribute the the genre rating. Corresponding holds for the actors and directors. the contribution item is considered a supporter if its contribution is strictly positive and a attacker if it is negative.

Item	Mean	Median	Standard deviation	Min	25%	75%	Max
Genre supporter	136.75	117.00	97.54	2.00	54.00	201.00	538.00
Genre attacker	83.26	68.00	66.13	0.00	31.00	119.00	657.00
Genre combined	220.01	198.00	149.50	4.00	90.00	316.00	902.00
Actor supporter	17.82	17.00	5.84	3.00	13.00	21.00	48.00
Actor attacker	7.49	6.00	3.46	0.00	5.00	9.00	38.00
Actor combined	25.32	24.00	7.19	4.00	20.00	29.00	59.00
Director supporter	4.80	4.00	3.74	2.00	3.00	5.00	57.00
Director attacker	1.42	1.00	1.50	0.00	1.00	2.00	35.00
Director combined	6.22	5.00	4.62	4.00	4.00	7.00	62.00

5.5 Distribution Of Item Strength

The statistic of the final strengths for the argumentation graphs as well as the strength from the arguments corresponding to the estimation of genre, actors and directors can be seen Table 4 with the corresponding distributions seen in Figure 20.

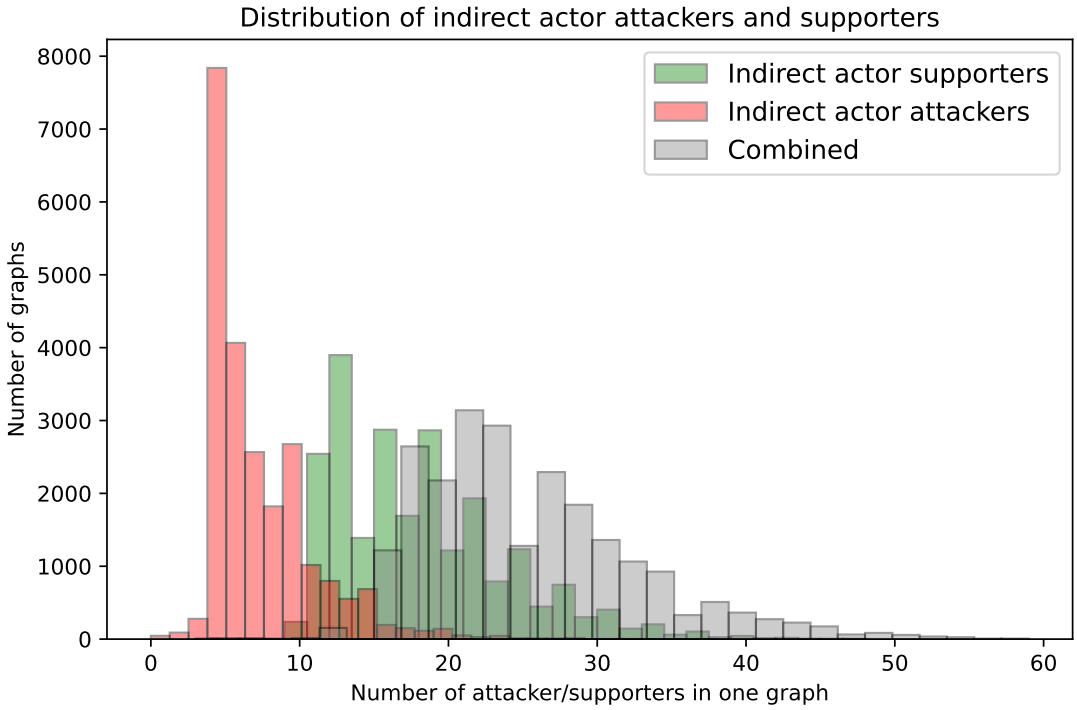


Figure 18: Shows the distribution of the attackers and supporters contributing to the rating for the actors. The contributing item is considered a supporter if its contribution is strictly positive and an attacker if it is negative.

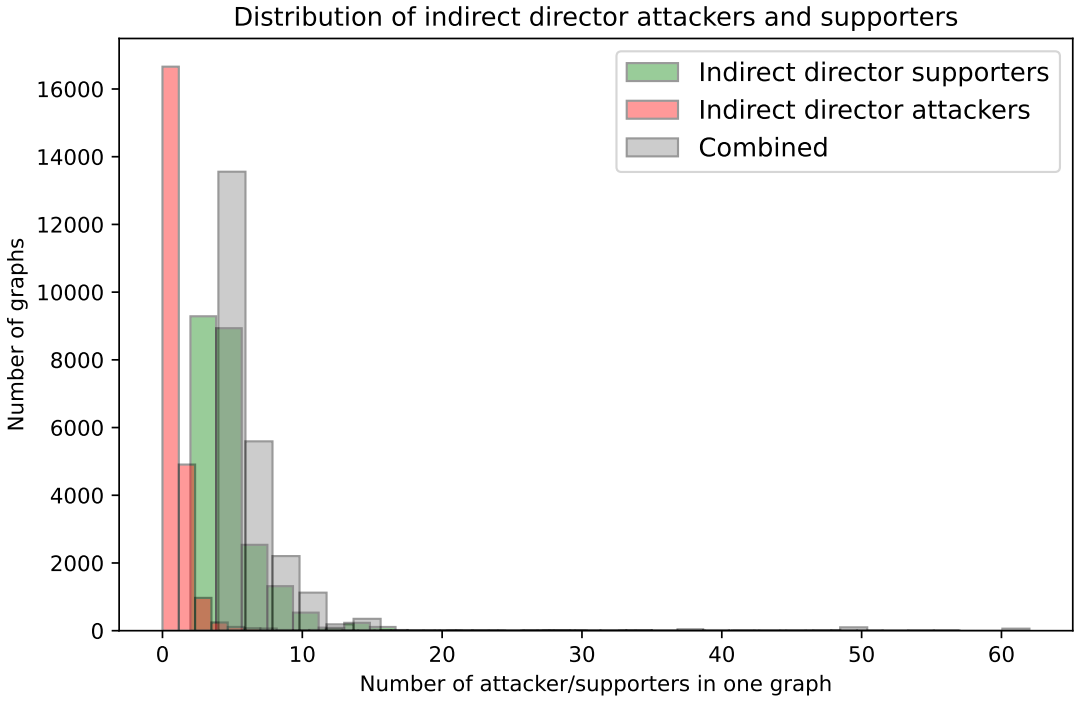


Figure 19: Shows the distribution of the attackers and supporters contributing to the rating for the director. The contributing item is considered a supporter if its contribution is strictly positive and an attacker if it is negative.

Table 4 Distribution of arguments strength among all 23,210 aggregated graphs based on the to last layers in the argumentation framework. The last layer contains the final strength and is the weighted average of the layer above with the rating for genre, actor and directors.

Item	Mean	Median	Standard deviation	Min	25%	75%	Max
Final	0.18	0.19	0.21	-0.71	0.05	0.33	0.89
Genre	0.22	0.23	0.26	-0.92	0.06	0.40	0.94
Actor	0.06	0.01	0.13	-0.93	0.00	0.10	1.00
Director	0.08	0.00	0.26	-1.00	0.00	0.02	1.00

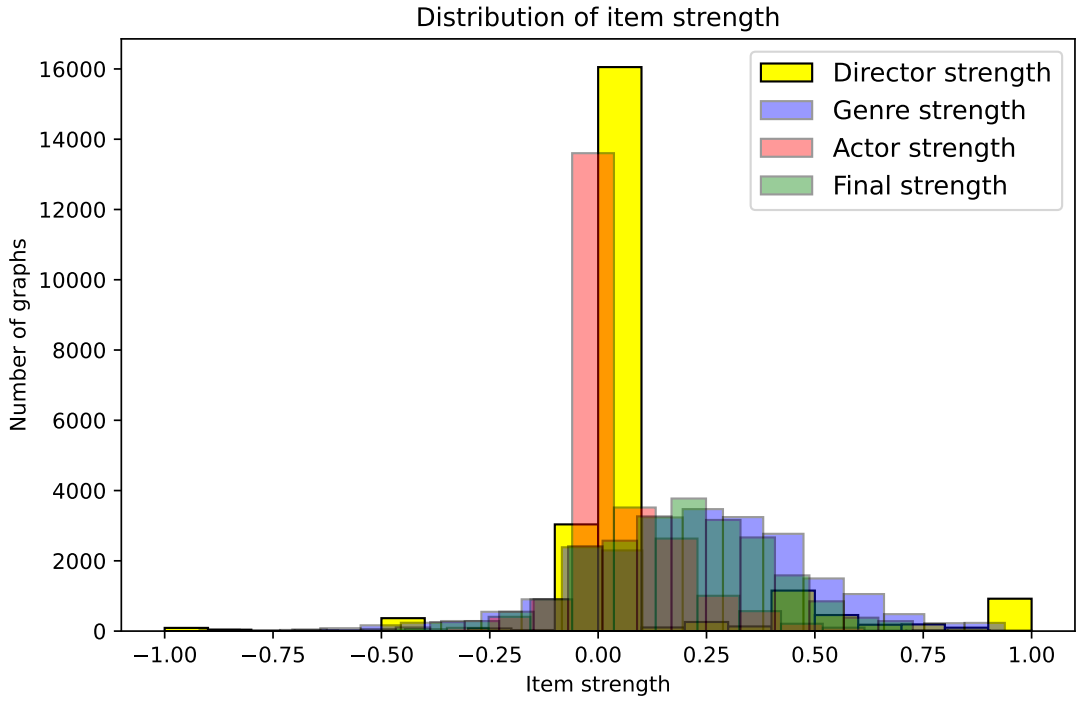


Figure 20: Shows the distribution of arguments strength among all 23,210 aggregated graphs based on the to last layers in the argumentation framework. The last layer contains the final strength and is the weighted average of the layer above with the rating for genre, actor and directors.

6 Discussion

This sections discuss the structures, data biases and distribution of different the properties of the graphs extracted in Section 5.

6.1 Acyclic Versus Cyclic Graphs

The graphs as can be seen in Figure 10 are acyclic by design so there exist no cyclic graphs in this argumentation framework. QBAFs in general can be cyclic so further studies need to be done on cyclic graphs using another argument framework.

6.2 Distribution Of The Graphs Input Ratings

As can be seen from the first layer for genre, actors and director in Table 1 it can be seen that the data used in the argumentation framework has different distributions for the genre, actors and directors. For the genre the median is 148.48 combined attackers and supporters 25% of the values are above 82 attackers and supporters. That means that for majority of the graphs there are a lot of data to generate the estimation for the generate strength. On other hand for the directors at least 25% of the graphs lacks either a attacker or supporter.

This will affect how the data are aggregate using Equation (3.1). Especially in the case where items are absent. In that case the framework will consider the aspect to do a neutral contribution. In the extreme case where there are no directors or actors like in Figure 15 the final strength will be given only by the genre. But it will not be given by the full genres strength but instead by the MUG multiplied by the genre strengthen that will in practice dampen the final strength.

Another approach would be to ignore contributions from aspects that that has low or no frequency. It is possible it could be handled by introducing a frequency dependent weight that would remove the contribution if there is no data instead of as now having the aspect contributing with a neutral strength. This would lead to the final strength in Figure 15 to be given by the full genre strength when the other aspects are absent. Investigating this further is outside the scope of this thesis and will be left to future work.

6.3 Strength Distributions

As can be seen in the Table 4 and Figure 20 the final strength corresponds much more to the genres strength than to the actors and directors strength. One reason for the is that the weight MUG is larger than for corresponding actor and director, so it is expected to contribute more. But it is also possible to see that although the max and min values for the genre, actor and director are similar the mean and median are not. One contributing factor to this is that at least 25% of the directors graphs are empty as can be seen from Table 1 and those contribute by moving the average to 0 instead of being excluded.

6.4 Bias

Another characteristic that is worth noting from Table 4 and from the initial layer 1 is that the graphs has more supporters than attacker leading to a bias towards a positive rating. The graphs in the data used has a positive bias in the initial rating that propagate to the final strength.

6.5 Graph Sizes

As can be seen from Table 2 the sizes or the graphs varies by a factor of $958/14 \approx 68$ between the smallest and largest graph. The fact that the smallest graphs are 14 is due the basic structure even without connecting ratings still has then nodes needed to potentially connect to ratings that don't exist. It can also be seen that most subgraphs for the actors and directors are small compared to the subgraphs generated from the genre. This corresponds to the observation of uneven distribution in 6.2 .

6.6 Relation To QBAFs

There is potential to use the framework to generate the initial weights for a QBAF where the attackers and supporters are given in accordance to the sign from the framework from QBAF. This investigation needs further theoretical work and is outside the scope of this paper.

7 Conclusion

The thesis has investigated the structure of the argumentation framework graphs used in [7] for a recommendation system. The graphs have been explicitly extracted and different properties analyzed. The graphs generated have a positive bias where most graphs are skewed to have more supporters than attackers in the initial data. The bias in the data propagates through the graph structure. The graphs generated are acyclic by the design of the framework. The result show that the structure of the ratings used for the different item types, genre, actor and directors are different. This has an impact on how the final strength, for instance when directors are missing the directors are set to contribute a neutral rating that in turn damped the final rating. More theoretical analysis is needed to understand how to properly translate the argumentation framework to a QBAF.

References

[1] Pietro Baroni, Antonio Rago, and Francesca Toni. From Fine-Grained Properties to Broad Principles for Gradual Argumentation: A Principled Spectrum. *International Journal of Approximate Reasoning*, 105:252–286, feb 2019.

[2] Cristian E. Briguez, Maximiliano C.D. Budán, Cristhian A.D. Deagustini, Ana G. Maguitman, Marcela Capobianco, and Guillermo R. Simari. Argument-based mixed recommenders and their application to movie suggestion. *Expert systems with applications*, 41(14):6467–6482, 2014.

[3] Timotheus Kampik, Nico Potyka, Xiang Yin, Kristijonas Čyras, and Francesca Toni. Contribution functions for quantitative bipolar argumentation graphs: A principle-based analysis. *International journal of approximate reasoning*, 173:109255–, 2024.

[4] Timotheus Kampik, Xiang Yin, Nico Potyka, and Francesca Toni. Strength change explanations in quantitative argumentation. AAMAS '26 (to appear). International Foundation for Autonomous Agents and Multiagent Systems, 2026.

[5] Timotheus Kampik, Kristijonas Čyras, and José Ruiz Alarcón. Change in quantitative bipolar argumentation: Sufficient, necessary, and counterfactual explanations. *International journal of approximate reasoning*, 164:109066–, 2024.

[6] Nico Potyka and Richard Booth. An empirical study of quantitative bipolar argumentation frameworks for truth discovery. In Chris Reed, Matthias Thimm, and Tjitze Rienstra, editors, *Computational Models of Argument - Proceedings of COMMA 2024, Hagen, Germany, September 18-20, 2024*, Frontiers in Artificial Intelligence and Applications, pages 205–216. IOS Press, 2024.

[7] Antonio Rago, Oana Cocarascu, Christos Bechlivanidis, and Francesca Toni. Argumentation as a Framework for Interactive Explanations for Recommendations. In *Proceedings of the 17th International Conference on Principles of Knowledge Representation and Reasoning*, pages 805–815, 9 2020.

[8] Antonio Rago, Hengzhi Li, and Francesca Toni. Interactive explanations by conflict resolution via argumentative exchanges. In Pierre Marquis, Tran Cao Son, and Gabriele Kern-Isberner, editors, *Proceedings of the 20th International Conference on Principles of Knowledge Representation and Reasoning, KR 2023, Rhodes, Greece, September 2-8, 2023*, pages 582–592, 2023.

[9] Paula Rodríguez, Stella Heras, Javier Palanca, Jhon M. Poveda, Néstor Duque, and Vicente Julián. An educational recommender system based on argumentation theory. *Ai communications*, 30(1):19–36, 2017.

[10] Cynthia Rudin. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature machine intelligence*, 1(5):206–215, 2019.

[11] Jose Ruiz Alarcon and Timotheus Kampik. Qbaf-py: Quantitative bipolar argumentation in c with python bindings. In *Computational Models of Argument*, pages 361–362. IOS Press, 2024.



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